

Hardware LLC associativity sweep - full results
(base kraken2, NO software cache, 50MB DB)

Reads	Assoc.	Cycles (M)	IPC	Wallclock	L2 hit%	NUCA hit%
10	1-way	13.6	1.64	115s	1.18%	0.0003%
10	4-way	13.7	1.64	117s	1.18%	0.0005%
10	8-way	13.6	1.64	117s	1.18%	0.0004%
10	15-way	13.7	1.64	116s	1.18%	0.0004%
10	16-way	13.7	1.64	114s	1.18%	0.0003%
10	30-way	13.7	1.64	116s	1.18%	0.0004%
50	1-way	24.4	1.47	193s	0.41%	0.0147%
50	4-way	24.4	1.47	194s	0.40%	0.0179%
50	8-way	24.4	1.47	193s	0.40%	0.0180%
50	15-way	24.4	1.47	193s	0.41%	0.0180%
50	16-way	24.4	1.47	193s	0.41%	0.0189%
50	30-way	24.4	1.47	194s	0.41%	0.0179%
100	1-way	870.0	1.58	1h 48m	0.21%	0.0332%
100	4-way	877.2	1.57	1h 48m	0.21%	0.0356%
100	8-way	869.9	1.58	1h 48m	0.21%	0.0361%
100	15-way	870.4	1.58	1h 48m	0.21%	0.0363%
100	16-way	869.8	1.58	1h 48m	0.21%	0.0364%
100	30-way	877.3	1.57	1h 47m	0.21%	0.0362%
500	1-way	4,528.2	1.59	9h 06m	0.20%	0.0316%
500	4-way	4,524.0	1.59	9h 13m	0.20%	0.0326%
500	8-way	4,537.9	1.59	9h 11m	0.20%	0.0336%
500	15-way	4,534.5	1.59	9h 05m	0.20%	0.0336%
500	16-way	4,526.4	1.59	9h 01m	0.20%	0.0335%
500	30-way	4,523.2	1.59	9h 09m	0.20%	0.0334%

24 runs: 6 real hardware LLC associativities (1/4/8/15/16/30-way) x 4 workload sizes (10/50/100/500 reads). Green = Luna's real 15-way. No software S2 cache in any row (base kraken2 binary throughout). Within each read-count group, cycles/IPC/hit-rates are flat - the null result: hardware associativity alone has no measurable effect here.